

Back paper Examination

Electrodynamics

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Duration: 120 minutes.

Total points: 60.

Please give arguments where necessary. If it is unclear from your answer why a particular step is being taken, full credit will not be awarded. Grades will be awarded not only based on what final answer you get, but also on the intermediate steps.

1. Suppose a charge density $\rho(\vec{r})$ be entirely contained in a finite spherical volume $\mathbb{V}$. Do not assume any symmetry of the charge distribution.
 - (a) Derive the relationship between the dipole moment of the sphere and the average of the electric field over the sphere.
 - (b) Is the spherical shape of $\mathbb{V}$ necessary for the result in the previous part? Explain why or why not.
 - (c) Consider a point dipole $\vec{p}$ at the origin. For **any** sphere that encloses the origin, the relationship you derived in the first part should be true. Argue that the far-field ($r \gg r'$) expression of the dipolar electric field (obtained by taking the gradient of $\frac{\vec{p} \cdot \vec{r}}{r^3}$) does not satisfy the result in part (a).
 - (d) Resolve the contradiction between the first and the third parts by supplementing (by hand) the electric field expression.

5 + 2 + 3 + 5 = 15 points

2. Consider three infinitely long, co-planar wires placed parallel to each other. Each carries the same current I in the same direction. The middle wire is placed at equal distance d from the other two.
 - (a) Find the location of the zeroes of the magnetic field.
 - (b) The middle wire is displaced a short distance x towards one of the wires, remaining parallel to its original orientation throughout the displacement process. Describe its subsequent motion in detail for $x \ll d$. Imagine the others are held in place.

- (c) Now the middle wire is displaced perpendicular to the common plane by the same distance x , again remaining parallel to its original orientation throughout. Like in the previous part, describe the subsequent motion of the middle wire. The other wires are always held in place.

5 + 5 + 5 = 15 points

3. (a) A toroidal coil has a uniform rectangular cross-section, with inner radius a , outer radius $a + w$, and height h . Consider the limit $w, h \ll a$. The toroid carries N turns of a current I which has a time-dependence $dI/dt = k$, k being a constant. Find the electric field at a point z above the center of the toroid.
- (b) A long solenoid of radius a and n turns per unit length carries a current $I(t)$ in the $\hat{e}_\phi$ direction. Find the electric field both outside and inside the solenoid.

8 + (3 + 4) = 15 points

4. (a) A particle with mass m and positive charge q is at rest at the origin at $t = 0$ in the presence of an electric field $\vec{E} = E_0 \hat{e}_z$ and a magnetic field $\vec{B} = B_0 \hat{e}_x$. Find the subsequent motion of the particle.
- (b) Find the velocity of the particle at **any** subsequent time t when its position coincides with the y -axis.

9 + 6 = 15 points